

Abstract Algebra - Stockholm University 2025

1. Groups

1.1. Conjugation, centralizer, center, normalizer

For $g \in G$, the conjugation map $c_g(x) = gxg^{-1}$ is an automorphism of G (an *inner* automorphism). For a subset $S \subseteq G$, the centralizer is

$$C_G(S) = \{ g \in G : gs = sg \text{ for all } s \in S \}; \quad C_G(g) := C_G(\{g\}).$$

The center is

$$Z(G) = C_G(G) = \{ g \in G : gx = xg \ \forall x \in G \}$$

(always a characteristic, hence normal, subgroup). For a subgroup $H \leq G$, the normalizer is

$$N_G(H) = \{ g \in G : gHg^{-1} = H \},$$

the largest subgroup of G in which H is normal (so $H \trianglelefteq N_G(H)$). One always has $C_G(H) \leq N_G(H)$. The number of conjugates of H in G equals $[G : N_G(H)]$.

1.2. Normal subgroups and quotients

A subgroup $N \leq G$ is normal (write $N \trianglelefteq G$) if $gNg^{-1} = N$ for all $g \in G$. Equivalently, every left coset equals the corresponding right coset: $gN = Ng$. If $N \trianglelefteq G$, the set of cosets $G/N = \{gN : g \in G\}$ is a group with $(gN)(hN) = (gh)N$.

1.3. Cosets and index

For $H \leq G$ and $g \in G$, the left (resp. right) coset is $gH = \{gh : h \in H\}$ (resp. $Hg = \{hg : h \in H\}$). Left cosets are either disjoint or equal and they partition G . Each left coset gH has $|H|$ elements via the bijection $H \rightarrow gH, h \mapsto gh$. The *index* of H in G is the number of left cosets, denoted $[G:H]$.

1.4. Lagrange's Theorem

If G is finite and $H \leq G$, then

$$|G| = [G:H] \cdot |H|.$$

Proof sketch. G is the disjoint union of $[G:H]$ left cosets, each of cardinality $|H|$.

Consequences

- $|H| \mid |G|$ for every $H \leq G$.
- For $g \in G$, the order $|g|$ divides $|G|$; hence $g^{|G|} = e$.
- Index multiplicativity: for $K \leq H \leq G$ (with G finite), $[G:K] = [G:H][H:K]$.
- If $[G:H] = 2$, then $H \trianglelefteq G$ (there are only two cosets, so gHg^{-1} must be H).

Groups of prime order are cyclic. If $|G| = p$ (prime) and $g \in G$, $g \neq e$, then $|\langle g \rangle| = |g| \mid p$ by Lagrange, so $|g| = p$ and $\langle g \rangle = G$. Hence G is cyclic and $G \cong \mathbb{Z}/p\mathbb{Z}$.

1.5. Isomorphism theorems

- (1) **First isomorphism theorem.** For $\varphi : G \rightarrow H$ a homomorphism,

$$G/\ker \varphi \cong \operatorname{im} \varphi.$$

- (2) **Second isomorphism theorem.** For $A \leq G$ and $N \trianglelefteq G$,

$$A \cap N \trianglelefteq A, \quad AN \leq G, \quad AN/N \cong A/(A \cap N).$$

- (3) **Third isomorphism theorem.** If $K \leq H \trianglelefteq G$, then $H/K \trianglelefteq G/K$ and

$$(G/K)/(H/K) \cong G/H.$$

- (4) **Correspondence (lattice) theorem.** For $N \trianglelefteq G$, the map

$$\{\text{subgroups } H \mid N \leq H \leq G\} \longleftrightarrow \{\text{subgroups of } G/N\}, \quad H \mapsto H/N$$

is a bijection preserving inclusion and index; it carries normal subgroups $H \trianglelefteq G$ with $N \leq H$ to normal subgroups of G/N .

2. Group actions

2.1. Actions on cosets and by conjugation

- For $H \leq G$, the action $G \curvearrowright G/H$ by left multiplication is transitive, with stabilizer of aH equal to aHa^{-1} .
- The action $G \curvearrowright G$ by conjugation, $g \cdot x = gxg^{-1}$, partitions G into conjugacy classes. The stabilizer of x is its centralizer $C_G(x)$, and $|\operatorname{Orb}(x)| = [G : C_G(x)]$.

2.2. Cayley's Theorem

Every group G is isomorphic to a subgroup H of the symmetric group $S_{|G|}$:

$$G \cong H \leq S_{|G|}.$$

2.3. Orbit–stabilizer

For any action $G \curvearrowright X$ and any $x \in X$ there is a canonical bijection

$$G/G_x \xrightarrow{\sim} \text{Orb}(x), \quad gG_x \mapsto g \cdot x.$$

In particular, if G is finite then

$$|\text{Orb}(x)| = [G : G_x] \quad \text{and} \quad |G| = |\text{Orb}(x)| \cdot |G_x|.$$

2.4. Class Equation

For a finite group G , let $Z(G)$ be the center of G . Then

$$|G| = |Z(G)| + \sum_{i=1}^r [G : C_G(g_i)],$$

where $g_1, \dots, g_r$ are representatives of the conjugacy classes not contained in $Z(G)$.

2.5. Conjugacy in S_n

Two permutations in S_n are conjugate if and only if they have the same cycle type (i.e. the same cycle lengths with the same multiplicities).

$$\sigma, \tau \in S_n \text{ are conjugate} \iff \text{cycle-type}(\sigma) = \text{cycle-type}(\tau).$$

Hence, the conjugacy classes of S_n correspond bijectively to partitions of n .

3. Sylows Theorem

4. Rings

5. Integral -, Unique Factorization - & Principle Ideal Domains

5.1. Gauss's Lemma

Let R be a UFD with field of fractions F , and let $p(x) \in R[x]$. If p is reducible in $F[x]$, then p is reducible in $R[x]$. (Equivalently: if p is irreducible in $R[x]$, then it is irreducible in $F[x]$.)

5.2. Eisenstein's Criterion

Let R be an integral domain and $P \subset R$ a prime ideal. Consider

$$f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in R[x].$$

If $a_{n-1}, \dots, a_0 \in P$ and $a_0 \notin P^2$, then $f(x)$ is irreducible in $R[x]$.

Corollary ($\mathbb{Z}[x]$). Let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0 \in \mathbb{Z}[x]$. If there exists a prime p such that $p \mid a_i$ for $i = 0, \dots, n-1$ and $p^2 \nmid a_0$, then f is irreducible in $\mathbb{Z}[x]$ (hence in $\mathbb{Q}[x]$ by Gauss).

6. Field Extensions